적분의 응용

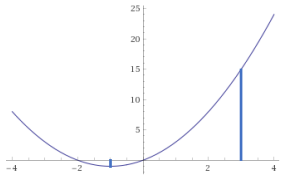
연습문제 풀이

08

연습문제 8.1

1. 주어진 구간 위의 f의 그래프와 x축으로 둘러싸인 영역의 면적을 구하여라.

(1) $f(x)=x^2+2x\,;[-1,3]$



$$\int_{-1}^{3}(x^2+2x)\,dx=\int_{-1}^{0}-(x^2+2x)\,dx+\int_{0}^{3}(x^2+2x)\,dx$$

$$A=-\left[\frac{1}{3}x^3\right]_{-1}^{0}-[x^2]_{-1}^{0}+\left[\frac{1}{3}x^3\right]_{0}^{3}+[x^2]_{0}^{3}=\left(-\frac{1}{3}+1+3^2+3^2\right)=\frac{56}{3}$$

(2) $f(x)=\cos x-\sin x\,;\left[0,\frac{\pi}{3}\right]$

$$\begin{pmatrix}0\le x\le\frac{\pi}{4},\ f(x)\ge 0\\ \frac{\pi}{3}\le x\le\frac{\pi}{4},\ f(x)\le 0\end{pmatrix}$$

$$\int_{0}^{\frac{\pi}{3}}(\cos x-\sin x)\,dx=\int_{0}^{\frac{\pi}{4}}(\cos x-\sin x)\,dx+\int_{\frac{\pi}{4}}^{\frac{\pi}{3}}-(\cos x-\sin x)dx$$

$$=[\sin x+\cos x]_{0}^{\frac{\pi}{4}}+[-\sin x-\cos x]_{\frac{\pi}{4}}^{\frac{\pi}{3}}$$

$$=\left(\frac{\sqrt{2}}{2}+\frac{\sqrt{2}}{2}-1\right)+\left(-\frac{\sqrt{3}}{2}-\frac{1}{2}+\frac{\sqrt{2}}{2}+\frac{\sqrt{2}}{2}\right)=2\sqrt{2}-\left(\frac{3+\sqrt{3}}{2}\right)$$

(3) $f(x)=\ln x\,;\left[\frac{1}{2},2\right]$

$$\begin{pmatrix}\frac{1}{2}\le x\le 0,\ f(x)\le 0\\ 0\le x\le 2,\ f(x)\ge 0\end{pmatrix}$$

$$\int_{\frac{1}{2}}^{2}\ln x\,dx=\int_{\frac{1}{2}}^{1}-\ln x\,dx+\int_{1}^{2}\ln x\,dx$$

$$=-[x\ln x-x]_{\frac{1}{2}}^{1}+[x\ln x-x]_{1}^{2}$$

$$=-\left(0-1-\frac{1}{2}\ln\left(\frac{1}{2}\right)+\frac{1}{2}\right)+(2\ln 2-2-0+1)$$

$$=\frac{3}{2}\ln 2-\frac{1}{2}$$

(4) $f(x)=\dfrac{x}{\sqrt{1-x^2}}$; $\left[-\dfrac{1}{2},\dfrac{1}{2}\right]$

$$\begin{pmatrix} -\dfrac{1}{2}\le x\le 0,\ f(x)\le 0 \\ 0\le x\le \dfrac{1}{2},\ f(x)\ge 0 \end{pmatrix}$$

$$\int_{-\frac{1}{2}}^{\frac{1}{2}}\frac{x}{\sqrt{1-x^2}}dx=\int_{-\frac{1}{2}}^{0}-\left(\frac{x}{\sqrt{1-x^2}}\right)dx+\int_{0}^{\frac{1}{2}}\frac{x}{\sqrt{1-x^2}}dx$$

$$\begin{pmatrix} u=1-x^2 & du=-2xdx \\ & xdx=-\dfrac{1}{2}du \end{pmatrix}$$

$$=\frac{1}{2}\int_{\frac{3}{4}}^{1}\frac{1}{\sqrt{u}}du-\frac{1}{2}\int_{1}^{\frac{3}{4}}\frac{1}{\sqrt{u}}du$$

$$=\frac{1}{2}\int_{\frac{3}{4}}^{1}\frac{1}{\sqrt{u}}du-\frac{1}{2}\int_{1}^{\frac{3}{4}}\frac{1}{\sqrt{u}}du$$

$$=\frac{1}{2}\left[2\sqrt{u}\right]_{\frac{3}{4}}^{1}-\frac{1}{2}\left[2\sqrt{u}\right]_{1}^{\frac{3}{4}}$$

$$=\frac{1}{2}\left(2-2\sqrt{\frac{3}{4}}\right)-\frac{1}{2}\left(2\sqrt{\frac{3}{4}}-2\right)=1-\frac{\sqrt{3}}{2}-\frac{\sqrt{3}}{2}+1$$

$$=2-\sqrt{3}$$

2.주어진 구간 위의 함수의 그래프 사이의 영역의 면적을 구하여라.

(3) $f(x)=\dfrac{1}{x}$, $g(x)=\dfrac{1}{x^2}$; $\left[\dfrac{1}{2},2\right]$

$$\text{교점}=\begin{pmatrix} \dfrac{1}{x}=\dfrac{1}{x^2} \\ x=1 \end{pmatrix}$$

$$\int_{\frac{1}{2}}^{2}[f(x)-g(x)]dx=\int_{\frac{1}{2}}^{2}\left(\frac{1}{x}-\frac{1}{x^2}\right)dx$$

$$=\int_{\frac{1}{2}}^{1}\left(-\frac{1}{x}+\frac{1}{x^2}\right)dx+\int_{1}^{2}\left(\frac{1}{x}-\frac{1}{x^2}\right)dx$$

$$=\left[-\ln x-\frac{1}{x}\right]_{\frac{1}{2}}^{1}+\left[\ln x+\frac{1}{x}\right]_{1}^{2}$$

$$=\left(-0-1+\ln\frac{1}{2}+2\right)+\left(\ln 2+\frac{1}{2}-0-1\right)$$

$$=0+\frac{1}{2}=\frac{1}{2}$$

(4) $f(x)=2x^3+2x^2$, $g(x)=2x^3-2x$; $[-4,2]$

$$\text{교점}=\begin{pmatrix} 2x^3+2x^2=2x^3-2x \\ 2x^2+2x=0 \\ 2x(x+1)=0 \\ x=-1\ \text{or}\ x=0 \end{pmatrix}$$

$$\int_{-4}^{2}[f(x)-g(x)]dx=\int_{-4}^{2}(2x^3+2x^2)-(2x^3-2x)dx$$

$$=\int_{-4}^{-1}(2x^3+2x^2)-(2x^3-2x)dx+\int_{-1}^{0}-(2x^3+2x^2)+(2x^3-2x)dx+\int_{0}^{2}(2x^3+2x^2)-(2x^3-2x)dx$$

$$=\left[\frac{2}{3}x^3+x^2\right]_{-4}^{-1}+\left[-\frac{2}{3}x^3-x^2\right]_{-1}^{0}+\left[\frac{2}{3}x^3+x^2\right]_{0}^{2}$$

$$=\left(-\frac{2}{3}+1+\frac{128}{3}-16\right)+\left(-\frac{2}{3}+1\right)+\left(\frac{16}{3}+4\right)$$

$$=27+\frac{1}{3}+\frac{28}{3}$$

$$=\frac{110}{3}$$

연습문제 8.2

1. 주어진 구간 위의 함수의 그래프와 x축 사이의 영역을 x축의 둘레로 회전시킨 입체 의 체적을 구하여라.

(2) $f(x)=\sqrt{2-x^2}$, $[0,\sqrt{2}]$

$$V=\pi\int_0^{\sqrt{2}}(\sqrt{2-x^2})^2dx=\pi\int_0^{\sqrt{2}}(2-x^2)dx=\pi\left[2x-\frac{1}{3}x^3\right]_0^{\sqrt{2}}$$

$$=(2\sqrt{2}-\frac{2}{3}\sqrt{2})\pi=\frac{4\sqrt{2}}{3}\pi$$

(3) $f(x)=\sin x$, $[0,\pi]$

$$V=\pi\int_0^{\pi}(\sin x)^2dx=\pi\int_0^{\pi}\sin^2 x dx=\pi\left[\frac{1}{2}x-\frac{1}{4}\sin(2x)\right]_0^{\pi}$$

$$=\frac{\pi^2}{2}$$

3. 주어진 구간 위의 함수의 그래프 사이의 영역을 x축 둘레로 회전시킨 입체의 체적을 구하여라.

(1) $f(x)=\sqrt{x+1}$, $g(x)=\sqrt{x-1}$ $[1,3]$

$$V=\pi\int_1^3(\sqrt{x+1})^2-(\sqrt{x-1})^2dx=\pi\int_1^3 2dx=\pi[2x]_1^3$$

$$=\pi(6-2)=4\pi$$

연습문제 8.3

1. 주어진 구간 위의 함수의 그래프와 x축 사이의 영역을 R이라고 할 때, R을 y축을 둘레로 회전시켜서 생긴 입체의 체적을 구하여라.

(3) $g(x)=\ln x,\ [1,3]$

$$V=2\pi\int_1^3 x(\ln x)dx=2\pi\left[\frac{1}{2}x^2\ln x-\frac{1}{4}x^2\right]_1^3$$

$$=2\pi\left(\frac{9}{2}\ln 3-\frac{9}{4}+\frac{1}{4}\right)=(9\ln 3-4)\pi$$

연습문제 8.4

1. 주어진 함수의 그래프의 길이를 구하여라.

(1) $f(x)=2x+3,\ [1,5]$

$$f'(x)=(2x+3)'=2$$
$$L=\int_1^5\sqrt{1+(2)^2}dx=\int_1^5\sqrt{5}dx=[\sqrt{5}x]_1^5$$

$$=5\sqrt{5}-\sqrt{5}=4\sqrt{5}$$

2. 곡선이 매개변수 t에 의하여 표현되어졌을 때 곡선의 길이를 구하여라.

(1) $x=t^2,\ y=2t+1,\ [0,\sqrt{3}]$

$$f'(t)=(t^2)'=2t$$
$$g'(t)=(2t+1)'=2$$
$$L=\int_0^{\sqrt{3}}\sqrt{4t^2+4}dt=2\int_0^{\sqrt{3}}\sqrt{t^2+1}dt$$

$$\left(\begin{array}{l} t=\tan u\quad dt=\sec^2 u du \\ \left|\begin{array}{l} t=0\ \rightarrow u=0 \\ t=\sqrt{3}\ \rightarrow u=\frac{\pi}{3}\end{array}\right.\end{array}\right)$$

$$=2\int_0^{\frac{\pi}{3}}\sec u\sec^2 u du=2\int_0^{\frac{\pi}{3}}\sec^3 u du=2\left[\frac{\sec u\tan u+\ln|\sec u+\tan u|}{2}\right]_0^{\frac{\pi}{3}}$$
$$=2\sqrt{3}+\ln|2+\sqrt{3}|$$

(3) $x=e^t\sin t,\ y=e^t\cos t,\ [0,\pi]$

$$f'(t)=(e^t\sin t)'=e^t\sin t+e^t\cos t$$
$$g'(t)=(e^t\cos t)'=-e^t\sin t+e^t\cos t$$
$$L=\int_0^{\pi}\sqrt{(e^t\sin t+e^t\cos t)^2+(-e^t\sin t+e^t\cos t)^2}dt$$
$$=\int_0^{\pi}e^t\sqrt{(\sin t+\cos t)^2+(-\sin t+\cos t)^2}dt$$
$$=\int_0^{\pi}e^t\sqrt{2+2\sin t\cos t-2\sin t\cos t}dt$$
$$=\int_0^{\pi}e^t\sqrt{2}dt=[\sqrt{2}e^t]_0^{\pi}=\sqrt{2}(e^{\pi}-1)$$

연습문제 8.5

2. 다음 극좌표로 표현된 점을 직교좌표로 바꾸어라.

(1) $\left(3, \frac{\pi}{4}\right)$

$x=(3)\cos\left(\frac{\pi}{4}\right)=\frac{3\sqrt{2}}{2}$

$y=(3)\sin\left(\frac{\pi}{4}\right)=\frac{3\sqrt{2}}{2}$

$\therefore\left(\frac{3\sqrt{2}}{2}, \frac{3\sqrt{2}}{2}\right)$

(4) $(5,0)$

$x=(5)\cos(0)=5$
$y=(5)\sin(0)=0$

$\therefore(5,0)$

(5) $\left(-2, \frac{\pi}{2}\right)$

$x=(-2)\cos\left(\frac{\pi}{2}\right)=0$

$y=(-2)\sin\left(\frac{\pi}{2}\right)=-2$

$\therefore(0,-2)$

3. 다음 직교좌표로 표현된 점을 극좌표로 바꾸어라.

(1) $(3,3)$

$r=\sqrt{3^2+3^2}=\sqrt{18}=3\sqrt{2}$
$\sin\theta=\frac{3}{3\sqrt{2}}, \cos\theta=\frac{3}{3\sqrt{2}}$ 이므로 $\theta=\frac{\pi}{4}$

$\therefore\left(3\sqrt{2}, \frac{\pi}{4}\right)$

(4) $(-4,0)$

$r=\sqrt{(-4)^2+0^2}=\sqrt{16}=4$
$\cos\theta=\frac{-4}{4}, \sin\theta=\frac{0}{4}$ 이므로 $\theta=\pi$

$\therefore(4,\pi)$

연습문제 8.6

1. 다음 직교 방정식을 극방정식으로 바꾸어라.

(1) $3x+2y=4$

$(3)r\cos\theta+(2)r\sin\theta=4$

(2) $x^2+9y^2=1$

$x^2+(3y)^2=1^2$
$(r\cos\theta)^2+(3r\sin\theta)^2=1$
$r^2\cos^2\theta+9r^2\sin^2\theta=1$
$r^2\cos^2\theta+9r^2\sin^2\theta-8r^2\sin^2\theta+8r^2\sin^2\theta=1$
$r^2(\cos^2\theta+\sin^2\theta)+8r^2\sin^2\theta=1$
$\therefore r^2+8r^2\sin^2\theta=1$

2. 다음 극방정식을 직교방정식으로 바꾸어라.

(1) $r=5$

$\sqrt{x^2+y^2}=5$
$\therefore x^2+y^2=25$

(2) $r=3\cos\theta$

$\sqrt{x^2+y^2}=3\cos\theta$
$\left(x=r\cos\theta \Rightarrow \cos\theta=\frac{x}{r}\right)$
$\sqrt{x^2+y^2}=\frac{3x}{\sqrt{x^2+y^2}}$
$x^2+y^2=3x$

$\therefore x^2-3x+y^2=0$